BTS –VI –04.16 – 0300 Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ]

A

B. Tech. Degree VI Semester Examination April 2016

CS 601 COMPILER CONSTRUCTION
(*2006 Scheme*)

Time: 3 Hours Maximum Marks: 100

PART A
(Answer *ALL* questions)

(8 × 5 = 40)

I. (a) Explain the role of lexical analyzer in a compiler.
(b) Write short notes on lexical analyzer tool LEX.
(c) Explain the role of a parser in a compiler.
(d) Compare and contrast recursive descent parser with predictive parser.
(e) Distinguish between synthesized and inherited attributes with examples.
(f) Explain activation records.
(g) Explain intermediate code generation.
(h) Explain peephole optimization.

PART B

(4×15 = 60)

II. What is a compiler? Explain different phases of a compiler with a neat diagram. (15)

OR

III. What is the significance of a transition diagram in the construction of lexical analyzer? Draw the transition diagram for (i) Relational operators (ii) Identifiers (iii) Numbers. (15)

IV. Compare Top Down parsing and Bottom Up parsing. Explain any Top Down parsing method. (15)

OR

V. Explain LR parsing algorithm. Construct LR parsing table for the grammar (15)

$E \longrightarrow E+T$

$E \longrightarrow T$

$T \longrightarrow T*F$

$T \longrightarrow F$

$F \longrightarrow (E)$

$F \longrightarrow id$

VI. Explain different storage allocation strategies. (15)

OR

VII. (a) Explain different parameter passing mechanisms. (10)
(b) Explain the use of symbol table. (5)

VIII. Explain different issues in the design of a code generator. (15)

OR

IX. What is three address codes? Explain the types of three address statements and their implementations. (15)

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